

CSE437 Data Science — Project Report

Cover

- **Project title:** Predicting Google Play Store App Ratings Using Machine Learning
- **Course:** CSE437 Data Science
- **Section:** »> 05 «<
- **Semester:** »> Summer 2026 «<
- **Group:** 15
- **Group members:**
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- **GitHub repository:** »><https://github.com/waysal1/cse437-google-play-app-rating-10>«<
- **Date:** September 3, 2026

Summary

This project predicts a Google Play Store app's user Rating (1–5 stars, continuous) from its listing metadata — category, review and install counts, size, price, type, and update recency — using the Kaggle *Google Play Store Apps* dataset (10,841 raw rows, cleaned to 8,196). We compared two regression families, Ridge Regression and Random Forest, after leakage-safe preprocessing, feature engineering, and hyperparameter tuning via cross-validation. Primary metric: test R^2 . Random Forest performed best ($R^2 = 0.164$, MAE = 0.338, RMSE = 0.499), narrowly ahead of Ridge ($R^2 = 0.140$); both clearly beat a mean-prediction baseline ($R^2 \approx 0$). The most important finding is that review volume and update recency carry the strongest, though still modest, signal, while error analysis shows the models fail specifically on low-review apps and at extreme ratings. This confirms the faculty's expectation that app metadata alone explains only a small share of rating variance, since it cannot capture app quality, user experience, or developer responsiveness — the deeper drivers of a subjective rating.

1. Problem and Dataset

1.1 Problem statement

Mobile application ratings are an important indicator of user satisfaction and app quality. However, ratings may be influenced by several characteristics such as app category, number of reviews, number of installs, size, price, and app type. The objective of this project is to analyze the factors associated with Google Play Store app ratings and develop machine-learning regression models to predict an application's rating based on its available characteristics.

1.2 Dataset

Source: Google Play Store Apps dataset (Kaggle) — <https://www.kaggle.com/datasets/lava18/google-play-store-apps> **License:** Creative Commons Attribution 3.0 Unported (CC BY 3.0), per the license.txt file distributed with the dataset download and reproduced at the repository root. **Collection method:** Apps and metadata scraped from the Google Play Store listing pages. **Size:** 10,841 rows $\times$ 13 columns as downloaded (googleplaystore.csv); after cleaning, 8,196 rows $\times$ 17 columns. **Time period:** No explicit collection date is provided; the Last Updated field ranges up to August 8, 2018, indicating the scrape was performed shortly after that date.

1.3 Target variable

Name: Rating. **Type:** continuous, nominally bounded 1.0–5.0. After cleaning, mean = 4.17, median = 4.3, std = 0.54 ($n = 8,196$). The distribution is strongly left-skewed: most apps sit between 4.0 and 4.5, with a long, sparse tail toward 1.0 (Figure 1).

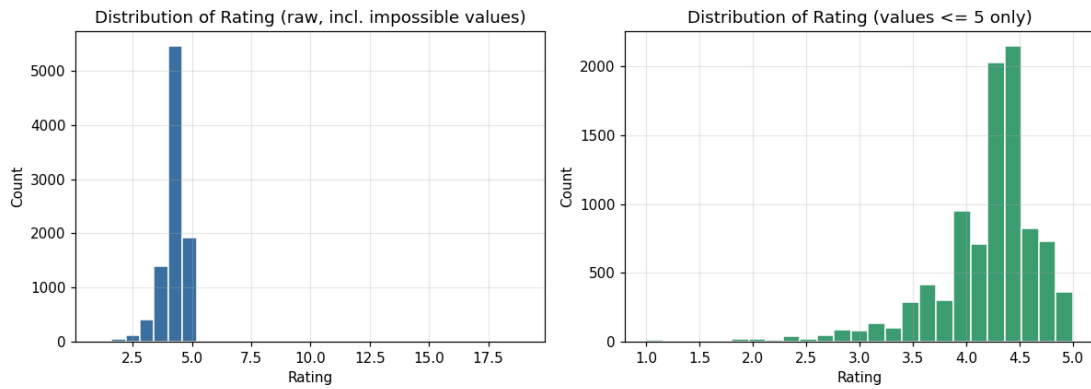


Figure 1: Rating distribution: raw (left, showing the single impossible value near 19) and valid range only (right)

1.4 Three questions

1. Which app characteristics, such as category, reviews, installs, size, price, and app type, are most strongly associated with Google Play Store ratings?
2. How accurately can machine-learning regression models predict an application's rating using its available characteristics?
3. Which machine-learning model and feature set provides the best rating prediction performance after preprocessing, feature selection, dimensionality reduction, and hyperparameter tuning?

2. Data Handling and Preprocessing

2.1 Data quality audit

- **Missing values:** Rating 1,474 (13.6%), Type 1, Content Rating 1, Current Ver 8, Android Ver 3.
- **Duplicates:** 9,660 unique App names out of 10,841 rows → 1,181 duplicate rows (some apps appear up to 9 times, e.g. ROBLOX).
- **Malformed row:** Row index 10472 (App = "Life Made WI-Fi Touchscreen Photo Frame") had every field from Category onward shifted one column left, producing an impossible Rating = 19.0. Reconstruction showed the true Category was blank at source, which caused the shift; all other fields were recoverable.
- **Inconsistent categories:** Category contained one non-category value, '1.9' — the fingerprint of the malformed row above.
- **Impossible values:** one Rating > 5 (the malformed row); a stray Type = '0' and Installs = '0'/'0+' values traced to the same row and to one newly-listed app with zero installs.

2.2 Missing values

Column	Strategy	Justification
Rating (target)	Drop row	A missing label cannot be used for supervised training; imputing a subjective rating would fabricate the exact quantity being predicted.
Category (malformed row only)	Drop row (1 row)	True value unrecoverable at source; negligible loss (0.009% of data).

Column	Strategy	Justification
Type, Content Rating	Verified — resolved automatically	Both single missing values coincided with rows already removed for missing Rating or the malformed row.
Current Ver, Android Ver	Retain as missing	Not named in the approved research questions; excluded from the model feature set rather than imputed.
Size (“Varies with device” → Size_MB missing, 1,170 rows)	Left missing; imputed only inside the training-only modelling pipeline (median)	Filling with a learned statistic before the train/test split would leak information; deferred per the leakage-prevention rule.

2.3 Outliers

Method: IQR rule ($[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$) applied to each numeric column on the cleaned dataset.

Column	% flagged	Decision
Reviews	~17%	Retain; log1p-transform (real right-skew, not error)
Installs	~24%	Retain; log1p-transform
Price	~7%	Retain, no transform (reflects the Free/Paid structural split)
Size_MB	~5.5%	Retain, no transform (plausible large-app sizes)
Rating	~6%	Retain (genuine low-rated apps; essential target signal)

No rows were removed for being statistically extreme; only genuinely invalid data (the malformed row, missing target) was removed.

2.4 Transformation and scaling

Raw text fields were parsed deterministically: Size → MB (M/k suffixes converted; “Varies with device” → missing), Installs → integer (comma/+ stripped), Price → float (\$ stripped), Reviews → integer, Last Updated → datetime. Category and Content Rating were one-hot encoded (handle_unknown='ignore', since two rare Content Rating levels occur only in the training split). Numeric features were standardized (StandardScaler) for Ridge only — Random Forest does not require scaling. **All imputation, scaling, and encoding was fit on the training split only**, inside sklearn Pipeline/ColumnTransformer objects, and applied unchanged to the test split.

2.5 Before and after

Stage	Rows	Cols	Unique apps	Duplicate rows	Missing Rating
Raw loaded	10,841	13	9,660	1,181	1,474
Malformed row handled	10,840	13	9,659	1,181	1,474

Stage	Rows	Cols	Unique apps	Duplicate rows	Missing Rating
Deduplicated (kept max-Reviews row)	9,659	13	9,659	0	1,463
Target validated (dropped missing Rating)	8,196	13	8,196	0	0
Parsed (Size_MB, Installs, Price, dates added)	8,196	17	8,196	0	0

3. Statistical Analysis

3.1 Descriptive statistics

Rating (n=8,196): mean 4.17, median 4.3, std 0.54. Reviews: heavily right-skewed, median 3,017 vs. mean 255,501. Installs: median 100,000 vs. mean 9.2 million — a handful of viral apps pull the mean far above the median. Paid apps (n=602, 7.3% of the data) average \$14.10. Size_MB averages 21.8 MB (median 13.0 MB) among the 7,026 apps with a known size. Category is dominated by FAMILY (1,651 apps) and GAME (898); Content Rating is dominated by Everyone (6,618, 81%); Type is 92.7% Free.

3.2 Relationships

Simple linear correlations with Rating: log_reviews 0.183 (strongest), days_since_update -0.130 (more recently updated apps rate higher), log_installs 0.085, Size_MB 0.063, Price_USD -0.021 (weakest). Category-level differences are real but mild — median Rating varies only from about 4.2 to 4.5 across the largest categories (Figure 2).

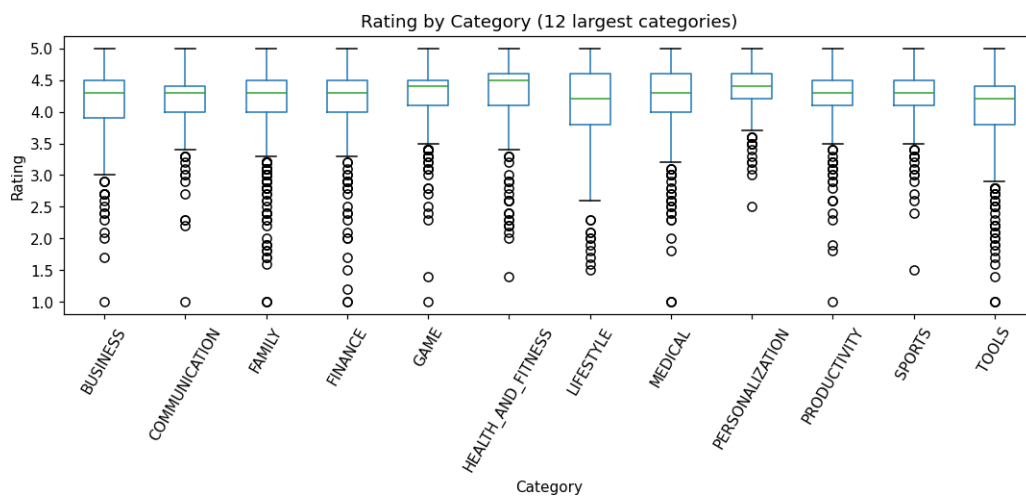


Figure 2: Rating by category, 12 largest categories

3.3 What the data says so far

1. The target is left-skewed and concentrated between 4.0–4.5, limiting how much variance any model can explain.
2. Even the best single predictor (log_reviews) correlates only weakly ($r \approx 0.18$) with Rating.
3. Category differences exist but are modest, not dominant.
4. Genres (114 levels) is largely a finer split of Category (33 levels) and adds cardinality without independent signal.
5. Several raw columns (Reviews, Size, Installs, Price) require explicit parsing before use.

4. Feature Engineering

4.1 Derived features

- log_reviews, log_installs — log1p of the two heavily right-skewed count columns.
- is_paid — binary flag from Price_USD > 0.
- days_since_update — chosen over a coarser last_updated_year after confirming both carry the same mirror-image signal; the finer-grained version was kept.
- Size_MB — reused from preprocessing.

4.2 Dimensionality reduction

PCA was fit on the training split only. Five components explain ~80% of variance, 9 components ~90%, 18 ~95% (Figure 3). **PCA was tested empirically via cross-validation and rejected:** it reduced Ridge's CV R^2 from 0.147 to 0.048, and Random Forest's from 0.130 to 0.096 (both fit on the same 100%-training-only comparison). PCA is **not** part of the final pipeline.

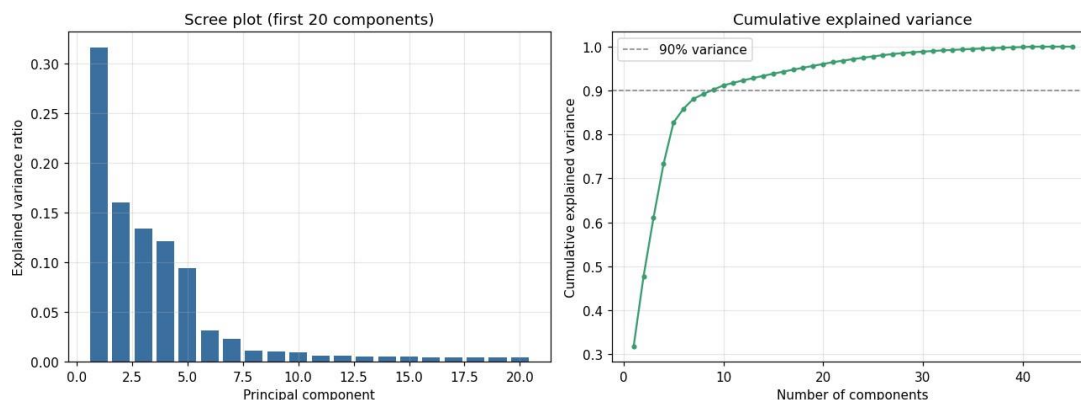


Figure 3: PCA scree plot and cumulative explained variance

4.3 Feature selection

Method: $f_{\text{regression}}$ via SelectKBest, fit on training data only. **Threshold:** $p < 0.05$. Of 45 one-hot-encoded feature dimensions, 22 were statistically significant; top scorers were log_reviews ($F=225.9$), days_since_update ($F=95.2$), Category_TOOLS ($F=50.5$), and log_installs ($F=46.6$). This subset was also tested empirically (CV $R^2 = 0.144$ for Ridge, vs. 0.147 for the full space) and showed **no measurable benefit**, so it was not adopted for final modelling either.

4.4 Final feature set

Kept (8 raw columns → 45 dimensions after one-hot encoding): log_reviews, log_installs, Size_MB, Price_USD, is_paid, days_since_update (numeric); Category, Content Rating (categorical).

Dropped: App (unique identifier); Type (redundant with is_paid); Genres (redundant with Category, 114 vs. 33 levels); Current Ver/Android Ver (near-unique version strings, not in the approved questions); raw Reviews/Installs/Price/Size strings and Last Updated/last_updated_year (superseded by parsed/derived versions).

5. Modeling and Validation

5.1 Validation strategy

An 80/20 train/test split (random_state=42) was performed **before** any learned transformation. Model development and tuning used 5-fold KFold(shuffle=True, random_state=42) cross-validation on the training split only. No temporal or grouping structure applies (each row is an independent app), so no special split logic beyond a fixed seed was needed.

5.2 Baseline

DummyRegressor(strategy='mean'): CV $R^2 = -0.0004 (\pm 0.0005)$, test $R^2 \approx 0.000$, MAE = 0.383, RMSE = 0.546 — the error scale any real model must beat.

5.3 Model families

Ridge Regression — suited to this problem because the one-hot-encoded feature space (45 dimensions, several correlated: e.g. log_reviews/log_installs) is moderately collinear; L2 regularization stabilizes coefficient estimates and yields interpretable, directly comparable coefficients. Assumes an approximately linear relationship between features and Rating, which caps achievable R^2 if the true relationship is non-linear.

Random Forest Regressor — suited because it makes no linearity assumption and can capture non-linear/interaction effects (e.g. category-specific review effects) without manual specification, and handles the mixed numeric/categorical space without scaling. Its main risk with a weak true signal is overfitting individual noisy splits; this is why max_depth, min_samples_split, and min_samples_leaf were tuned.

5.4 Metrics

Primary metric: R^2 , declared before any test result was seen, because it is scale-free, comparable across model families, and the metric the faculty's expected range (0.05–0.15) is stated in. **Secondary metrics:** MAE, RMSE, reported in rating-point units for practical interpretability.

6. Hyperparameter Tuning

6.1 Search space

Model	Hyperparameter	Values searched
Ridge	alpha	13 log-spaced values, 0.001 to 1000
Random Forest	n_estimators	100, 150, 200, 250
Random Forest	max_depth	5, 10, 15, 20, None
Random Forest	min_samples_split	2, 5, 10
Random Forest	min_samples_leaf	1, 2, 4
Random Forest	max_features	sqrt, log2, 0.5

6.2 Method

Ridge: exhaustive GridSearchCV, 13 candidates $\times$ 5 folds = 65 fits, scoring = r2. **Random Forest:** the full grid (540 combinations) was not computationally reasonable on the single-CPU environ-

ment used, so RandomizedSearchCV was used instead: 20 candidates $\times$ 5 folds = 100 fits, scoring = r^2 .

6.3 Results

Ridge: best $\alpha = 10$, $CV R^2 = 0.1475$. $CV R^2$ is essentially flat across a wide middle range of α and only degrades at extremes (Figure 4) — the ceiling is set by weak signal, not regularization strength.

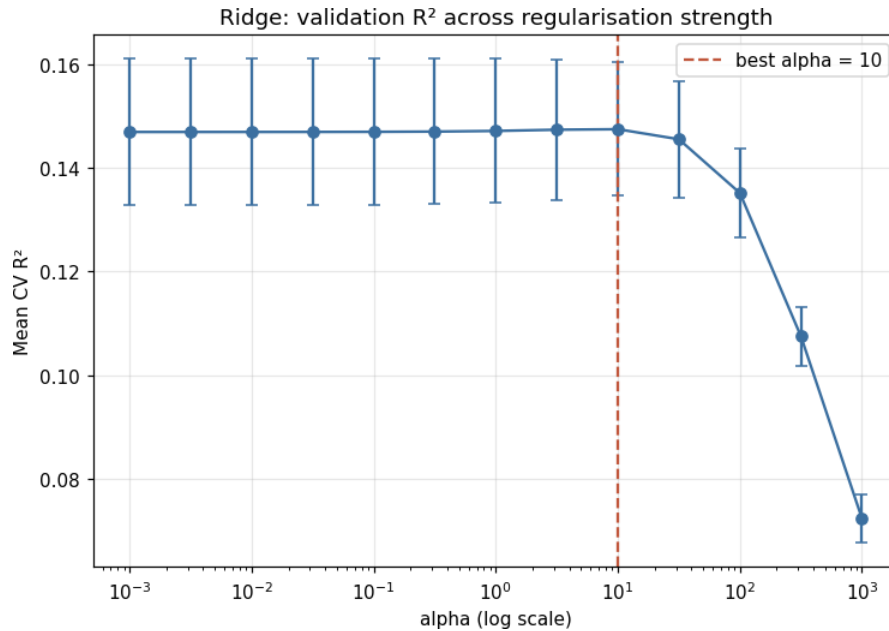


Figure 4: Ridge validation R^2 across regularization strength

Random Forest: best configuration $n_estimators=200$, $max_depth=None$, $min_samples_split=10$, $min_samples_leaf=2$, $max_features=0.5$, $CV R^2 = 0.1533$ (up from 0.1138 with default hyperparameters — a real tuning gain, unlike Ridge). Very shallow trees ($max_depth=5$) clearly underperformed; depth stopped mattering much beyond 10–20 (Figure 5).

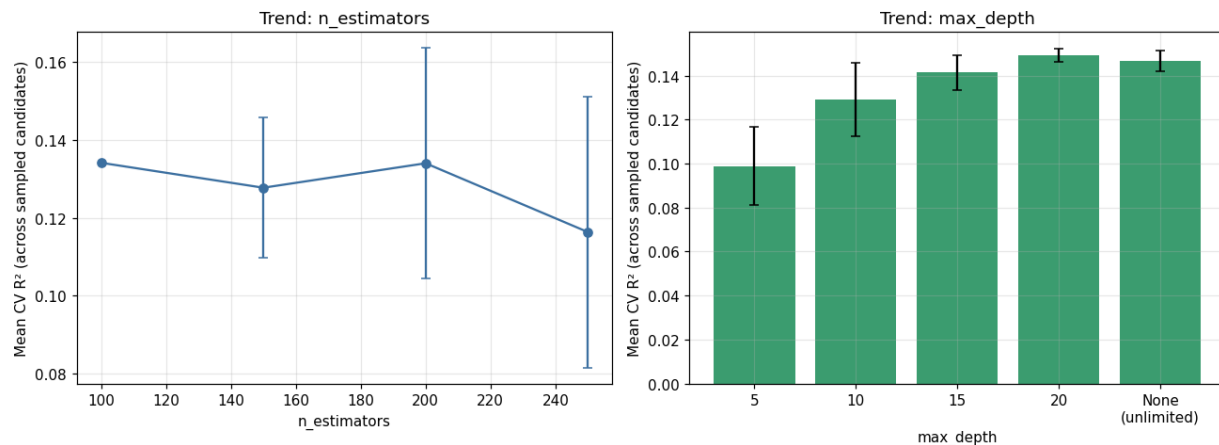


Figure 5: Random Forest tuning trends: $n_estimators$ and max_depth

7. Results, Visualization and Error Analysis

7.1 Test set performance

Evaluated once, after all model development:

Model	R^2	MAE	RMSE
Dummy baseline	~ 0.000	0.383	0.546
Ridge (tuned)	0.140	0.345	0.506
Random Forest (tuned)	0.164	0.338	0.499

7.2 Visualization

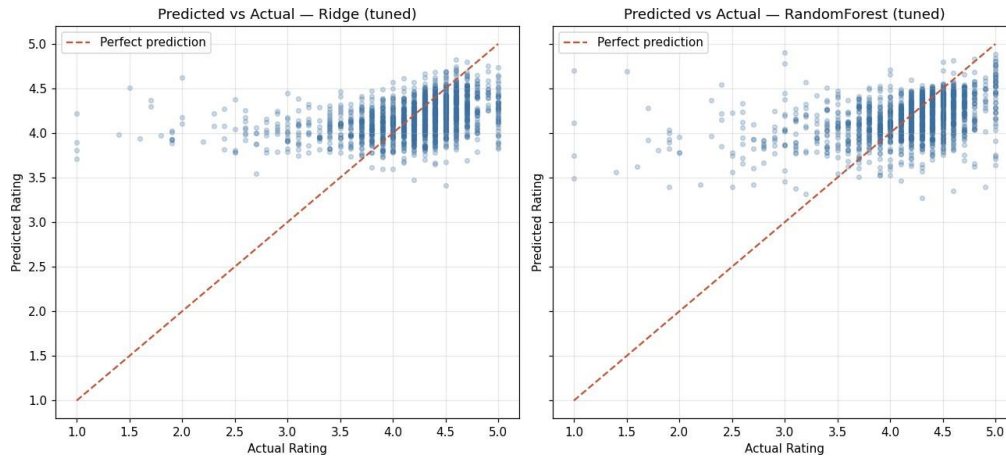


Figure 6: Predicted vs actual, both models

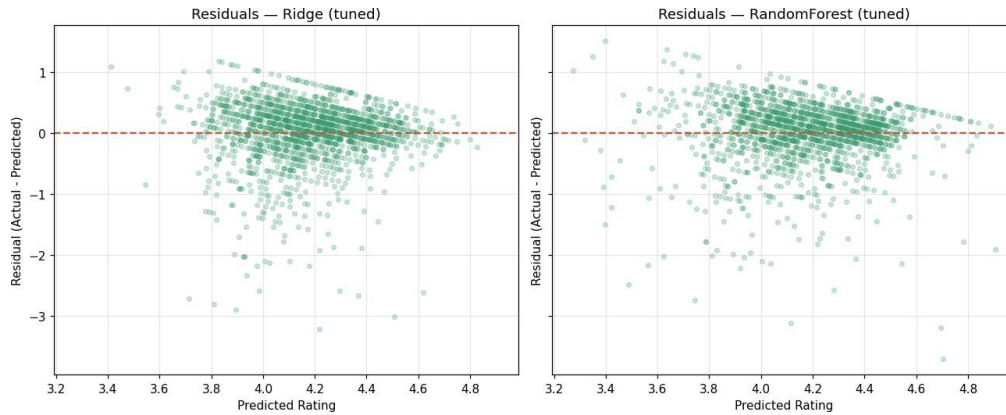


Figure 7: Residuals vs predicted, both models

Both models' predictions cluster in a narrow band (~ 3.8 – 4.5) regardless of the true rating (Figure 6), and residuals show a long tail of large negative values — true low-rating apps the models failed to identify (Figure 7). RandomForest importance (Figure 8) is dominated by `log_reviews` (0.274), `days_since_update` (0.188), `Size_MB` (0.164), and `log_installs` (0.130); Category dummies contribute smaller individual amounts. Ridge's largest-magnitude coefficients are `log_reviews` (+0.58) and `log_installs` (−0.51); this opposite-sign pair is a collinearity artifact (the two are highly correlated), not evidence that installs hurt ratings — RandomForest importance, unaffected by this sign-splitting, shows both contributing positively. The error-distribution

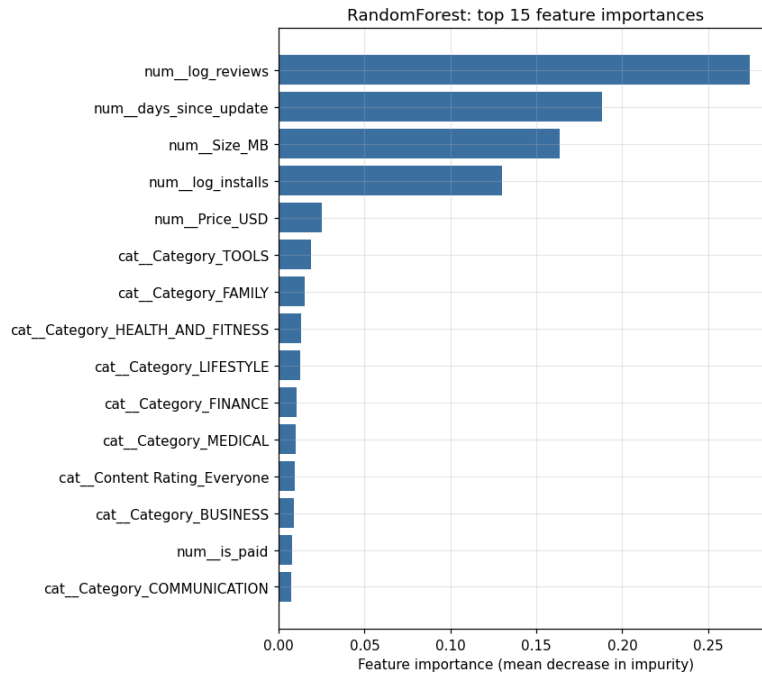


Figure 8: RandomForest feature importance, top 15

histogram and full Ridge coefficient chart are saved in figures/13_error_distribution.png and figures/15_ridge_coefficients.png.

7.3 Error analysis

Subgroup	MAE (Random Forest)
Reviews < 100	0.548
Reviews ≥ 100	0.271
Installs ≥ 1,000,000	0.183
Installs < 1,000,000	0.385
Free apps	0.337
Paid apps	0.357
Rating 1.0–2.0	2.371
Rating 4.0–4.5	0.169

Errors are roughly **double** for low-review apps and **highest by far** at extreme low ratings — a clear regression-to-the-mean pattern: with little usage history to go on, the model defaults toward the population mean (~ 4.17), so genuinely low-rated apps are predicted far too high. DATING (MAE 0.646) and LIFESTYLE (0.511) are the worst-predicted categories; EDUCATION (0.182) and SHOPPING (0.217) the best.

Example 1 — “House party - live chat” (DATING): actual rating 1.0, predicted 4.70 (error -3.70); only 1 review and 10 installs — almost no usage signal, so the model falls back near the mean, while the true low rating likely reflects a specific bad experience the metadata cannot capture.

Example 2 — “FK Atlanta” (SPORTS): actual rating 1.5, predicted 4.70 (error -3.20); only 2 reviews and 5 installs — the same mechanism: too little signal to distinguish this app from any other unrated SPORTS app.

7.4 Answers to the three questions

Q1 — Which characteristics matter most? Review volume (`log_reviews`) and update recency (`days_since_update`) are the two strongest associations by a clear margin across every method used (correlation, `f_regression`, Random Forest importance, Ridge coefficients). `log_installs` and `Size_MB` are moderately important. Category matters diffusely (many small individual effects, no single dominant category). `Price/is_paid` show the weakest, though still statistically detectable, association.

Q2 — How accurately can the models predict rating? Modestly. Tuned models reach test $R^2 = 0.140\text{--}0.164$, $\text{MAE} \approx 0.34\text{--}0.35$, $\text{RMSE} \approx 0.50\text{--}0.51$ stars — a real improvement over the baseline ($\text{MAE } 0.383$), but the models are typically off by roughly a third of a star, and far less accurate for low-review apps and rating extremes (Section 7.3).

Q3 — Which model/feature set performs best? Tuned Random Forest on the **full 45-dimension feature space** ($R^2 = 0.164$), narrowly ahead of tuned Ridge ($R^2 = 0.140$). Both PCA and `f_regression` feature selection were tested and **reduced** performance, so neither is part of the best configuration. The margin between the two model families is narrow enough that both should be read as extracting a similarly modest amount of real signal.

8. Limitations and Next Steps

App metadata carries only a **weak** signal for a subjective quantity like Rating. Major likely drivers of a real user's rating are absent from this dataset: functional quality (bugs, crashes, load times), user-experience design, advertising load, developer responsiveness, evolving user expectations, and privacy concerns (plausibly relevant to the worst-predicted categories, `DATING` and `HEALTH_AND_FITNESS`). The dataset is also a single scrape, not a time series, so only associations — not causal or trend claims — are supported. With more time or data, useful extensions include: NLP features from the companion user-review text/sentiment file, developer-level reputation or update-history data, and category-specific models given how much error varies by category. The faculty's expected R^2 range ($0.05\text{--}0.15$) was met or nearly met without leakage or forced tuning; these results are an honest ceiling for metadata-only prediction, not a modelling shortfall.

9. Contributions

Member	Student ID	Contribution
Khondokar Waysal E Mustafa	23341001	All stages: data audit, preprocessing, feature engineering, modelling, tuning, evaluation, error analysis, and report writing.

References

- Dataset: Lavanya, *Google Play Store Apps*, Kaggle. <https://www.kaggle.com/datasets/lava18/google-play-store-apps> (CC BY 3.0).
- Libraries: pandas, NumPy, scikit-learn, matplotlib.
- AI assistance disclosure:** Claude (Anthropic) was used throughout this project to help write and execute the Jupyter notebooks (data auditing, preprocessing, feature engineering, modelling, tuning, and evaluation code), to design the analysis pipeline in line with the faculty's leakage-prevention and methodology requirements, and to draft this report from the notebooks' actual outputs. All code was executed and all reported numbers were verified against real notebook output before inclusion; no results in this report were fabricated or estimated.